

Project 1: Data Warehousing & ETL Architecture

🔗 **Parent Page:** SQL Masterclass: Data Engineering & Analytics

> 💡 **Project Objective:**
> Design and execute an Extract, Transform, Load (ETL) pipeline using Microsoft SQL Server to integrate raw CSV data from CRM and ERP systems into a centralized Star Schema using the Medallion Architecture

https://github.com/Youssef-Hany1/sql-etl-data-warehouse

🧠 Core Architecture (The Medallion Approach)

- 🥉 **Bronze Layer:** Raw data ingestion. Focuses on bringing CRM and ERP data into the database using `BULK INSERT` and `TRUNCATE` operations without applying any transformations to maintain traceability.
- 🥈 **Silver Layer:** Cleansed and standardized data. Focuses on handling missing data (`COALESCE`), standardizing domains (e.g., gender and marital status mapping), casting data types, and generating technical metadata columns (e.g., load dates).
- 🥇 **Gold Layer:** Business-ready Star Schema. Highly refined data integrating multiple sources into Dimension tables (Customers, Products) using Surrogate Keys, and a central Fact table (Sales) exposed via SQL `VIEWS`.

🛠️ Technical Implementation

- **Database Objects:** Stored Procedures, Views, Star Schema Data Modeling.
- **SQL Commands:** Complex DDL/DML, `TRY...CATCH` for error handling, dynamic execution duration logging, and `ROW_NUMBER()` for surrogate key generation.

Tables

DWH Phases

Aa Phases	Progress
Requirement Analysis	1
Design Data Architecture	1
Project Initialization	1
Build Bronze Layer	1
Build Silver Layer	1
Build Gold Layer	1

DWH Tasks

Phases	Tasks	Checkbox
Requirement Analysis	Analyze & Understand The Requirements	☑
Design Data Architecture	Choose Data Management Approach	☑
Design Data Architecture	Design The Layers	☑
Design Data Architecture	Draw The Data Architecture (Draw.io)	☑
Project Initialization	Create Detailed Project Tasks (Notion)	☑
Project Initialization	Define Project Naming Conventions	☑
Project Initialization	Create GIT Repo & Prepare The Repo Structure	☑
Project Initialization	Create Database & Schemas	☑
Build Bronze Layer	Analyzing: Source Systems	☑
Build Bronze Layer	Coding: Data Ingestion	☑
Build Bronze Layer	Validating: Data Completeness & Schema Checks	☑
Build Bronze Layer	Document: Draw Data Flow (Draw.io)	☑
Build Bronze Layer	Commit Code in Git Repo	☑
Build Silver Layer	Analyzing: Explore & Understand Data	☑
Build Silver Layer	Document: Draw Data Integration (Draw.io)	☑
Build Silver Layer	Coding: Data Cleansing	☑
Build Silver Layer	Validating: Data Correctness Checks	☑
Build Silver Layer	Documenting: Extend Data Flow (Draw.io)	☑
Build Silver Layer	Commit Code in Git Repo	☑
Build Gold Layer	Analyzing: Explore Business Objects	☑
Build Gold Layer	Coding: Data Integration	☑
Build Gold Layer	Validating: Data Integration Checks	☑
Build Gold Layer	Document: Draw Data Model of Star Schema (Draw.io)	☑
Build Gold Layer	Document: Create Data Catalog	☑
Build Gold Layer	Document: Extend Data Flow (Draw.io)	☑
Build Gold Layer	Commit Code in Git Repo	☑